

Google's data center energy use doubled in 4 years

No wonder Google is desperate for more power: The company's data centers more than doubled their electricity use in just four years.

The eye-popping stat comes from Google's [most recent sustainability report](#), which it released late last week. In 2024, Google data centers used 30.8 million megawatt-hours of electricity. That's up from 14.4 million megawatt-hours in 2020, the earliest year Google broke out data center consumption.

Google has pledged to use only carbon-free sources of electricity to power its operations, a task made more challenging by its breakneck pace of data center growth.

And the company's electricity woes are almost entirely a data center problem. In 2024, data centers accounted for 95.8% of the entire company's electron budget.

Google's Electricity Use (MWh)

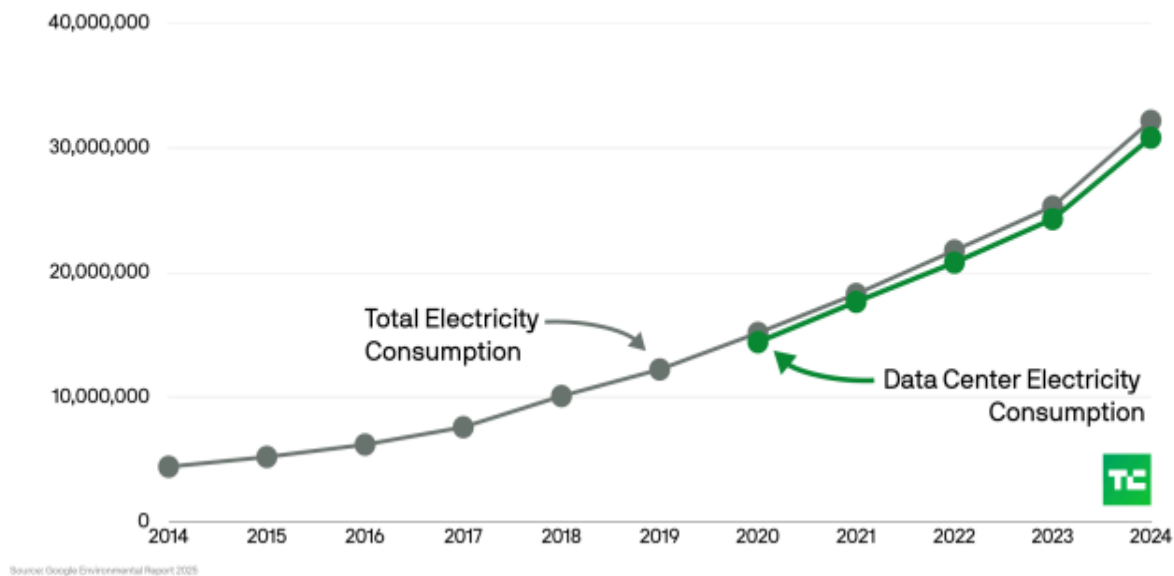


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The company's ratio of data-center-to-everything-else has been remarkably consistent over the last four years. Though 2020 is the earliest year Google has made data center electricity consumption figures available, it's possible to use that ratio to extrapolate back in time. Some quick math reveals that Google's data centers likely used just over 4 million megawatt-hours of electricity in 2014. That's sevenfold growth in just a decade.

The tech company has already picked most of the low-hanging fruit by improving the efficiency of its data centers. Those efforts have paid off, and the company is frequently lauded for being at the leading edge. But as [the company's power usage effectiveness](#) (PUE) has approached the theoretical ideal of 1.0, progress has slowed. Last year, Google's company-wide PUE dropped to 1.09, a 0.01 improvement over 2023 but only 0.02 better than a decade ago.

It's clear that Google needs more electricity, and to keep to its carbon-free pledge, it has been investing heavily in a range of energy sources, including geothermal, both flavors of nuclear power, and renewables.

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Geothermal shows promise for data center operations. By tapping into Earth's heat, enhanced geothermal power plants can consistently generate electricity regardless of the weather. And many startups, including [Google-backed Fervo Energy](#), are making it possible to drill profitable wells in more places.

On the nuclear fusion side, Google last week announced it would [invest in Commonwealth Fusion Systems](#) and buy 200 megawatts of electricity from its forthcoming Arc power plant, scheduled to come online in the early 2030s. In the nuclear fission world, Google has pledged to buy 500 megawatts of electricity [from Kairos Power](#), a small modular reactor startup.

The nuclear deals have yet to deliver power — and they won't for five years or more. In the meantime, the company has been on a renewable energy buying spree. In May, the company bought [600 megawatts of solar capacity](#) in South Carolina, and in January, it announced a deal for [700 megawatts of solar in Oklahoma](#). Google said in 2024 it was working with Intersect Power and TPG Rise Climate to build [several gigawatts' worth](#) of carbon-free power plants, a \$20 billion investment.

The outlay isn't surprising given that solar and (to a lesser extent) wind are the only two sources of power that are readily available before the end of the decade.

New nuclear power plants take years to permit and build, and even the most optimistic timelines don't see them connecting to the grid or a data center before the end of the decade. Natural gas, which the U.S. has plenty of, is hamstrung by five-plus-year waitlists for new turbines. That leaves renewables paired with battery storage.

Google has contracted with enough renewables to match its total consumption, though those sources don't always deliver electrons when and where the company needs them.

"When we announced to the world that we would achieve that 100% annual matching goal, we were very clear that wasn't the end state," Michael Terrell, Google's head of advanced energy, told reporters last week. "The end game was 24/7 carbon-free energy around the clock everywhere we operate at all times."

Google has some work to do.

Worldwide, the company has about 66% of its data center consumption, matched to the hour, powered by carbon-free electricity. But that average papers over some regional challenges. While its Latin American data centers hit 92% last year, its Middle East and Africa facilities are only at 5%.

Those hurdles are part of why Google is investing in stable, carbon-free sources like fission and fusion, Terrell said. "In order for us to eventually reach this goal, we are going to have to have these technologies," he said.

Topics

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